

Inequalities

(P10). Trichotomy S has a subset

$$P \subseteq S \rightarrow \text{"}P \text{ is a subset of } S\text{"}$$

exactly one of the following is true:

①. $a = 0$

②. $a \in P$

"add.
inv." $\leftarrow$ ③. $-a \in P$

Ex. ① If $S = \mathbb{Z}$ (integers)

We can take $P = \{1, 2, 3, \dots\}$

But $P' = \{-1, -2, -3, \dots\}$

$P'' = \{1, -2, 3, -4, \dots\}$

②. In $S = \mathbb{Z}/11\mathbb{Z} = \{1, 2, 3, \dots, 11\}$

We could take $P = \{1, 2, 3, 4, 5\}$

③. In $S = \mathbb{Z}/12\mathbb{Z}$, P10 cannot hold: $6 = -6$

$\Rightarrow$ ② ③ always hold simulta.
for $a = 6$.

(P11). Closure under addition.

If $a, b \in P$, then so is $a+b$.

(Then, in $S = \mathbb{Z}$, can't take $P = \{1, -2, 3, -4, \dots\}$ etc.)

(P12). Closure under multiplication

If $a, b \in P$, then so is $a \cdot b$.

(Then in $S = \mathbb{Z}$, can't take $P = \{-1, -2, \dots\}$)

Then we can rule out $\mathbb{Z}/11\mathbb{Z}$

pf. if $3 \in P$, so is $3+3=6$. ($5 \notin P$)

($8 \notin P$)

$3 \cdot 3 = 9$. ($2 \notin P$)

Then $6+6=12 \in P$ ($10 \notin P$)

But then $1+1=2 \in P$ & Both are contradictions
 $1+9=10 \in P$ $\downarrow$

Def For $a, b \in P$

• $a > b$ if $a + (-b) \in P$

• $a \geq b$ if $a + (-b) \in P$
or $a + (-b) = 0$.

Lemma If $a \neq 0$
then $a^2 > 0$

PF If $a \in P$.

$$\begin{aligned}\text{By } p12 \quad a \cdot a &= a^2 \in P \\ &= a^2 + (-0) \\ &\Rightarrow a^2 > 0\end{aligned}$$

If $-a \in P$.

$$\begin{aligned}\text{By } p12, \quad (-a)(-a) &\in P \\ \parallel \\ a \cdot a &= a^2\end{aligned}$$

So $a^2 > 0$.



Def For $a \in S$, the abs. value of a , denoted as $|a|$, is

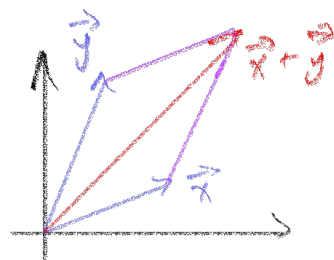
$$|a| = \begin{cases} 0, & \text{if } a = 0 \\ a, & \text{if } a \in P \\ -a, & \text{if } -a \in P \end{cases}$$

↓
By Trichotomy, one of these is always true.

Thm. (Triangle Inequality)

For all $a, b \in S$,

we have $|a+b| \leq |a| + |b|$



PF. ① if $a, b \geq 0$

$$|a+b| = \begin{cases} \bullet \text{ if } a=0, b \neq 0 \\ |a+b| = |0+b| = |b| = a+b \quad (b \in p) \\ \bullet \text{ if } a \neq 0, b=0 \quad \text{* same as above,} \\ \text{just switch } a \text{ and } b \\ \bullet \text{ if } a=b=0, \\ |a+b| = 0 = a+b \\ \bullet \text{ if } a, b \neq 0 \\ \text{then } a, b \in p \xRightarrow{\text{CPII}} a+b \in p \\ \Rightarrow |a+b| = a+b \end{cases}$$

In this case, $|a|=a, |b|=b$, then $|a+b| \leq |a| + |b|$ is equivalent to $a+b \leq a+b$, which is true.

② $a \geq 0, b < 0$ ($b \geq 0, a < 0$)

• If $a+b \geq 0$, then

first write $b = -c$ ($c \in P$)

$$|a+b| = |a-c| = a-c$$

$$|a| + |c| = a+c$$

$$\text{Then } (a+c) - (a-c) = c+c \in P \quad (P'')$$

• If $a+b \notin P$, ^{$\cup \{0\}$} then $|a+b| = -(a+b)$

$$|a| = a$$

$$|b| = -b$$

$$|a| + |b| - |a+b|$$

$$= a - b - (- (a+b))$$

$$= a+a$$

$$= 2a \in P \cup \{0\}$$

$$\text{so } |a| + |b| \geq |a+b|$$

③. If $a, b \notin P \cup \{0\}$.